

Ethan Carter Edwards

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Education

A.B. in Honors Computer Science, Harvard University

(August 2025 - May 2029)

- 3.8 GPA; Coursework: Grad. Distributed Systems, Grad. Systems Security, Operating Systems, Intro Systems.

Experience

Teaching Fellow for CS 61 Systems Programming, Harvard University

(August 2026 - December 2026)

Infrastructure Software Engineer, Exa.ai

(May 2026 - Current)

- Architected a monorepo build system to manage complex acyclic dependencies between 500+ projects with Nix.
- Designed and implemented a Kubernetes plugin to replace OCI images with Nix manifests, accelerating deployments 5x.
- Sped up build and CI times by 10-100x through leveraging and writing custom multi-derivation Nix builders and tooling.
- Implemented an AWS S3-backed caching server for Nix builds and Nix flake inputs using Go, Cloudflare, and k8s.
- Gave two presentations at DEFCON Nix Vegas demonstrating how we use Nix to model and manage our infrastructure.

Summer of Nix Fellow, NixOS Foundation

(June 2025 - September 2025)

- Used Nix, NixOS, Rust, Python, and OCaml to support 50+ programs and modules for clients across the E.U.

IT Support Intern Team Lead, Carroll County Public Schools

(March 2023 - August 2025)

- Managed a team of 6 interns across three buildings over the span of three years. Hired interns each year.
- Wrote PowerShell scripts to simplify software delivery process for district, saving hundreds of man-hours.
- Deployed 1,200+ Chromebooks and 80+ AV systems; conducted professional development training to faculty and staff.

Systems Administrator Intern, AlthaTechnology

(March 2020 - August 2021)

- Designed and implemented a distributed storage system serving up to 30 servers with 10Gb/s fiber network.
- Implemented distributed and highly available compute layer using Docker Swarm, reducing costs by \$300/month.

Projects and Activities

Homelab, Open-Source NixOS/Nixpkgs work, ZFS, IaC, Networking, Docker, Kubernetes, C++, Nix

- Maintain a multi-architecture web-hosting fleet to deploy sites with 2K+ daily active users at a 99% uptime.
- Hardware includes Dell Poweredge R320, Cisco 3750G L3 Switch, and managed WiFi Access Points.
- Serve on Nix MacOS, CUDA, NGI, Tmux teams. Maintain 500+ packages. Review 2500+ PRs. 450+ PRs merged.

The Ridge Podcast, NextJS, React, Flask, Cloudflare R2, SCSS, Docker, REST APIs

[Website](#)

- Developed a custom podcast CMS to serve 500+ daily listeners with 99% uptime using a global CDN system.
- Interviewed 100+ politicians, historians, veterans, musicians, journalists, writers, teachers, and more.

The Harvard Crimson Tech Board, Django, React, Cloudflare R2, GraphQL

[Website](#)

- Migrated site serving 21 million annual viewers from Amazon S3 to Cloudflare R2, saving \$400/month.
- Debugged cache invalidation issues, developed interactive article editor, ArchieML parser.

Dracula Tmux, Bash, REST APIs, Open-Source

[850+ GitHub Stars](#)

- Maintainer for a popular Open-Source modular tmux plugin for Linux and *BSD systems.
- Developed statuses for Kubernetes, Git, Terraform, Weather, GPU/RAM/CPU utilization, and Blood Glucose levels.

Traxos, Multi-Paxos Distributed BitTorrent Tracker

- Fully-featured BitTorrent Tracker written in C++ using the Multi-Paxos Distributed Consensus Algorithm.

Certifications

- [FCC Amateur Radio Amateur Extra Class License](#)
- [CompTIA A+ \(1101-1102\)](#)
- [CompTIA Security+](#)
- [\(ISC\)² \(CC\)](#)